ELEC 271 Digital Systems
Fall 2026

Tutorial Week 2
Conversion Between Binary and Decimal Systems

1.1 Convert the following decimal numbers into binary, using the method shown in Figure 1.6.
(a) $(20)_{10}$
(b) $(100)_{10}$
(c) $(129)_{10}$
(d) $(260)_{10}$
(e) $(10240)_{10}$

Solution:

1.1. (a) 10100 (b) 1100100 (c) 10000001 (d) 100000100 (e) 10100000000000

1.6 Convert the following binary numbers into decimal.
(a) $(1001)_2$
(b) $(11100)_2$
(c) $(111111)_2$
(d) $(101010101010)_2$

Solution:

1.6. (a) 9 (b) 28 (c) 63 (d) 2730

1.8 What is the minimum number of bits needed to represent the following decimal numbers in binary?
(a) $(270)_{10}$
(b) $(520)_{10}$
(c) $(780)_{10}$
(d) $(1029)_{10}$

Solution:

1.8. (a) 9 (b) 10 (c) 10 (d) 11